

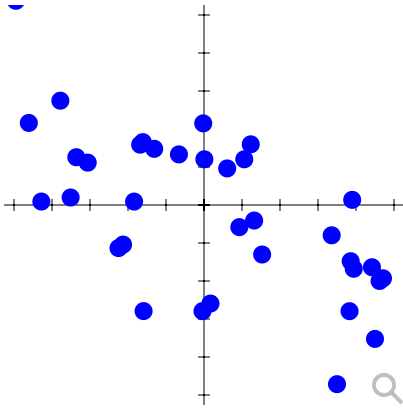
Exam 3 Review

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Question 1

0/1 pt 3 19

Here is a scatter plot for a set of bivariate data.



What would you estimate the correlation coefficient to be?

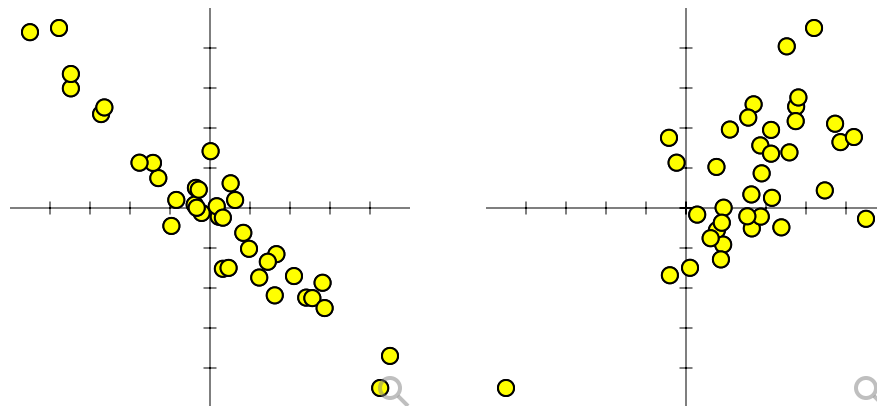
-0.9 -0.6 0 0.6 0.9

☐ ☐ ☐ ☐ ☐

Question 2

0/1 pt 3 19

Here are the scatter plots for two sets of bivariate data with the same response variable. The first compares the variables x & y . The second compares the variables w & y



Which explanatory variable has a stronger relationship with the response variable y ?

- ☐ The first variable (x) has a stronger relationship with the response variable y .
- ☐ The second variable (w) has a stronger relationship with the response variable y .

Question 3

0/1 pt 3 19

Based on the data shown below, a statistician calculates a linear model

$$y = 2.87x + 3.75.$$

x	y
4	15.5
5	17.85
6	19.6
7	25.45
8	27
9	28.95
10	32.4

Use the model to estimate the y-value when $x = 4$

y = _____

Use the model to estimate the y-value when $x = 6$

y = _____

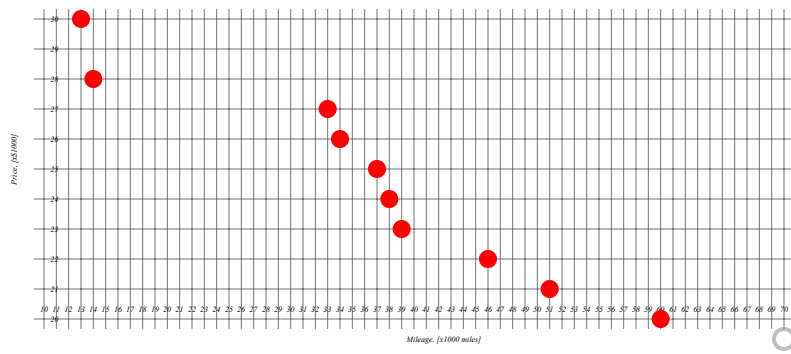
● Question 4

0/1 pt 3 19

Consider the following data set that contains information about a sample of ten 2014 Subaru Outback offered for sale.

Vehicle	Condition	Color	Mileage (x1000miles)	Price (x\$1000)
1	average	red	13	30
2	average	white	14	28
3	good	black	33	27
4	average	silver	34	26
5	poor	green	37	25
6	average	red	38	24
7	poor	white	39	23
8	poor	red	46	22
9	poor	black	51	21
10	average	green	60	20

The scatter plot that summarizes the data with regard to mileage as the input variable and price as the output variable is provided below.



a. Is there an association between the two variables? If yes, is it positive or negative?

There appears to be between the two variables.

b. Use technology to find the equation of the line of best fit, $y = mx + b$, and the coefficient of determination and interpret the slope, the vertical intercept, and the coefficient of determination. (Round the answers to 3 decimal places.)

i. $m =$ is the ;

ii. $b =$ is the ;

iii. $r^2 =$ hence there is a linear relation.

c. Use the line of best fit to:

i. estimate the price of a car with 64500 miles.

Answer: _____ dollars (Round the answer to the nearest unit.)

ii. estimate the mileage of a car whose price is 21500 dollars.

Answer: _____ miles (Round the answer to the nearest unit.)

● Question 5

0/1 pt 3 19

Use the equation $U = -0.1M - 0.2$ to

i. find the output U when the input $M = -2$.

Answer: _____ (Round the answer to 2 decimal places)

ii. find the input M when the output $U = 0$.

Answer: _____ (Round the answer to 2 decimal places)

● Question 6

0/1 pt 3 19

Fill in the missing values for this ANOVA source table, and use it to calculate the effect size:

	<i>SS</i>	<i>df</i>	<i>MS (S²)</i>	<i>F</i>
Between	319.92	6		
Subject	5124.672	18		
Error				
TOTAL	12140.592			

$R^2 =$ _____

● Question 7

0/1 pt 3 19

Below is the raw data for an experiment. Conduct the appropriate ANOVA hypothesis test with $\alpha=0.01$, calculating the appropriate bracketed terms, and filling in the source table.

If decimals exceed 3 places, round to 3 places.

Participant	Black	G1	G2	G3	White
A	5	4	6	6	6
B	5	4	5	4	5
C	1	5	8	5	4
D	3	8	10	4	4

To get $[Y]$, square each term, then add them up all up: $[Y] =$ _____

For $[A]$, add up each column, then square those numbers: $[A] =$ _____

For $[S]$, you follow the same process, but flip your rows and columns: $[S] =$ _____

For $[T]$, add up every datapoint, square that number, and divide by the total number of datapoints.: $[T] =$ _____

Here's some cool formulas:

$$SS_{Bet} = [A] - [T]$$

$$SS_{Sub} = [S] - [T]$$

$$SS_{Tot} = [Y] - [T]$$

Source of Variation	SS	df	MS (S^2)	F
Between Groups	_____	_____	_____	_____
Subjects	_____	_____	_____	_____
Error	_____	_____	_____	
Total	_____	_____		

$F_{crit} =$ _____

Based upon the results of your ANOVA, do you reject, or fail to reject the null hypothesis?

- ☐ Do not reject
- ☐ Reject

What is the effect size?

$R^2 =$ _____

● Question 8

☑ 0/1 pt ↻ 3 ⇌ 19

What is the F_{crit} value with for a within groups study with 5 groups, having 10 participants total, and a level of significance (α) of 0.05?

If using software, round to two decimal places.

$F_{crit} =$ _____

● Question 9

☑ 0/1 pt ↻ 3 ⇌ 19

A dean wants to check the consistency of teaching quality among the faculty in one of the departments. A sample of students was obtained from five classes taught by five different faculty, and their common final exam scores were recorded.

Classes	Final Exam Scores									
Class 1	71	72	83	69	75					
Class 2	71	76	73	87	75	65	70	62	63	
Class 3	73	67	69	91	64	70	69	74	71	89
Class 4	67	73	46	57	74	71				
Class 5	67	77	77	87	65	72	66			

Can you conclude that the teaching quality is inconsistent among the courses? Use the level of significance 1%.

Procedure:

Assumptions: (select everything that applies)

- ☐ Normal populations
- ☐ All expected frequencies are 1 or greater
- ☐ At most 20% of the expected frequencies are less than 5
- ☐ Independent samples
- ☐ Simple random samples
- ☐ Equal population standard deviations